PathologyLoss Figure Atlas

Every figure produced by the 50-epoch publication run on BraTS 2021 — twenty-two plates, in the order the argument needs them, each marked for whether it can go in the paper as it stands.

RUN	SUBJECTS	SPLIT
phasefinal_seed1	100	80 / 20 by patient
EPOCHS	**OBJECTIVES**	**WALL CLOCK**
50	5 + 2 anchors	36 h 05 m

CITE usable as it stands CARE usable with a caveat stated

RETIRE contradicts the current analysis

ACT 01

The problem

Rician noise at σ = 0.08, applied in the image domain the way magnitude-reconstructed MRI actually carries it. Segmenting the corrupted input scores 0.3186 Dice on enhancing tumour; segmenting the clean image scores 0.6345. Everything in this study lives in that gap.

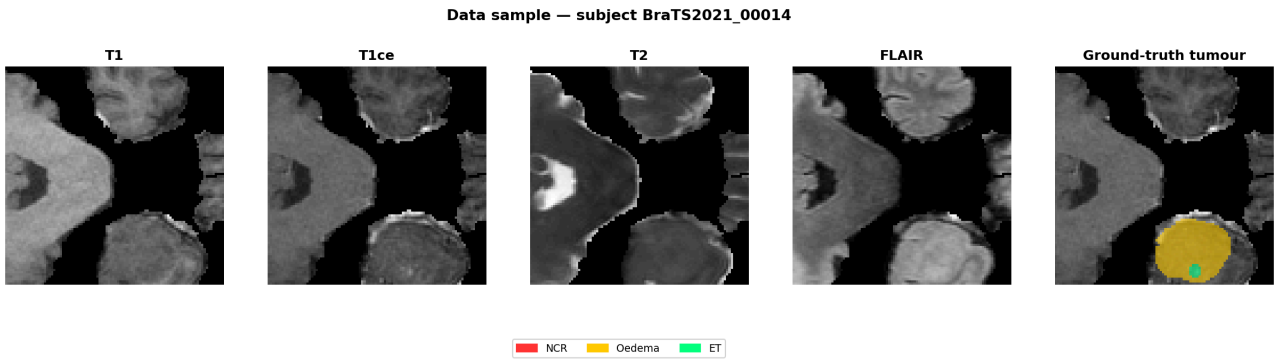
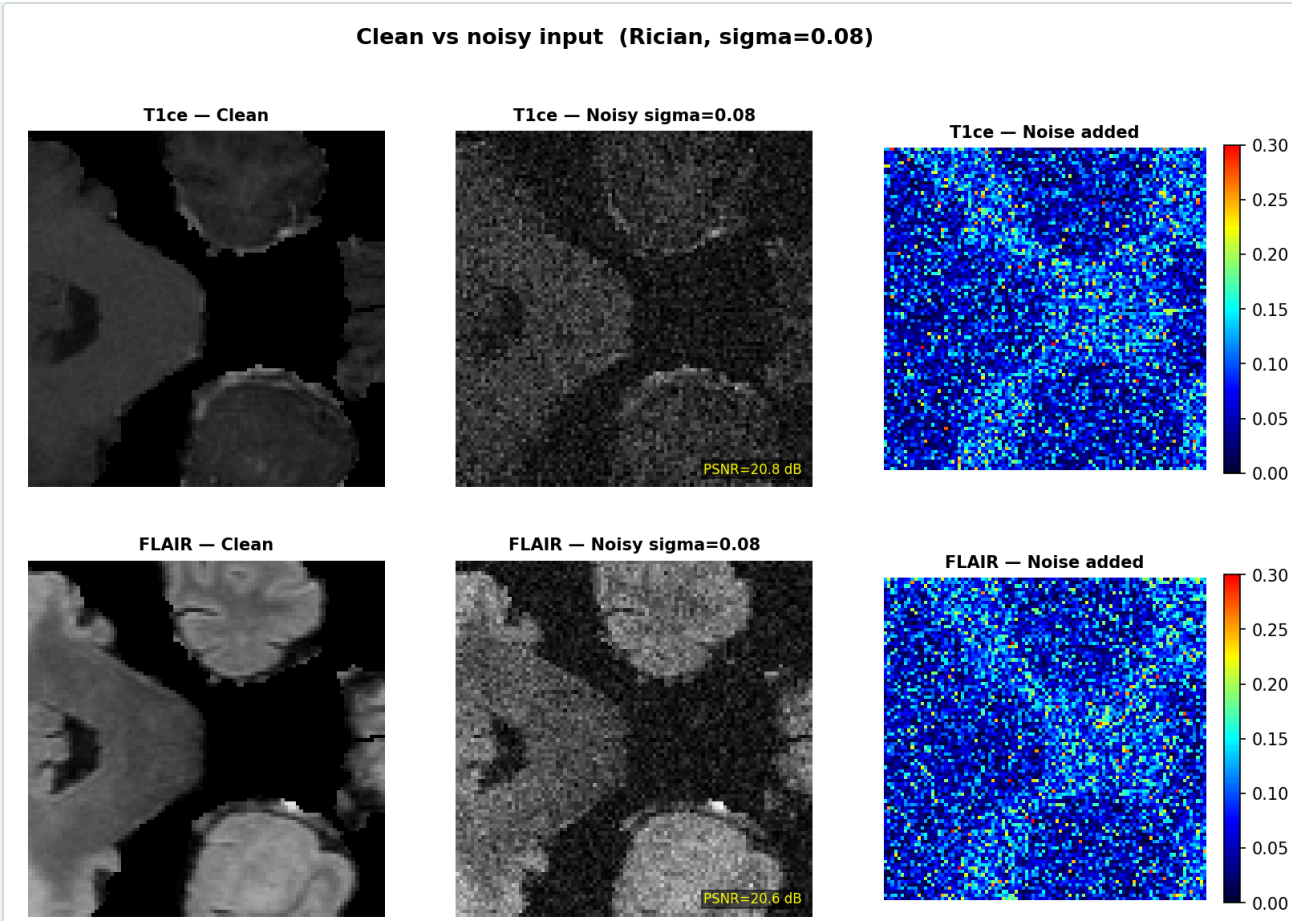


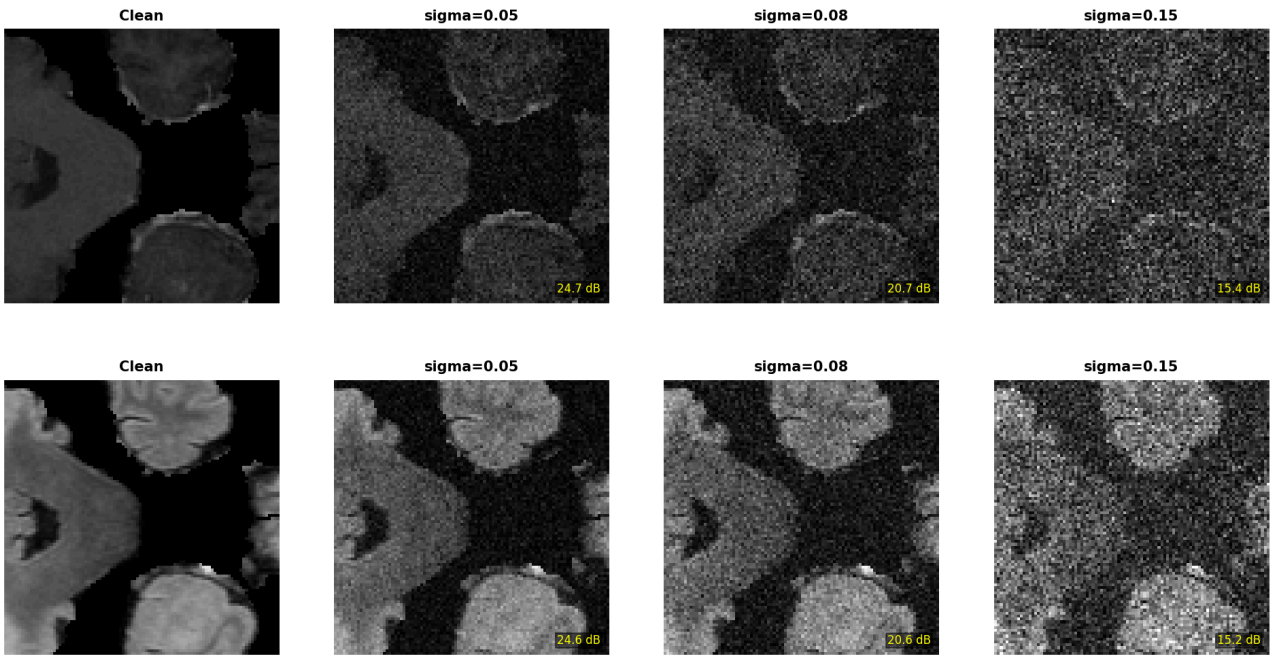
fig01 The input data CITE

One subject, four co-registered modalities — T1, T1ce, T2, FLAIR — with the expert label map. T1ce is where enhancing tumour shows, which is why every qualitative plate here uses it.



`fig02` **What the corruption does** CITE

Clean against corrupted, same windowing. The damage is not uniform in consequence: fine enhancing-tumour margins disappear while bulk white matter survives looking acceptable.



`fig10` **Noise progression** CITE

The same slice across increasing σ, showing where 0.08 sits. Useful for justifying the operating point — though note the study only ever trains and tests at 0.08, so generalisation across this axis is untested.

ACT 02

The method

A restoration network trained with an extra term that weights reconstruction error inside the nested BraTS regions — whole tumour ⊃ tumour core ⊃ enhancing tumour — at 1 : 2 : 3, each normalised by its own area.

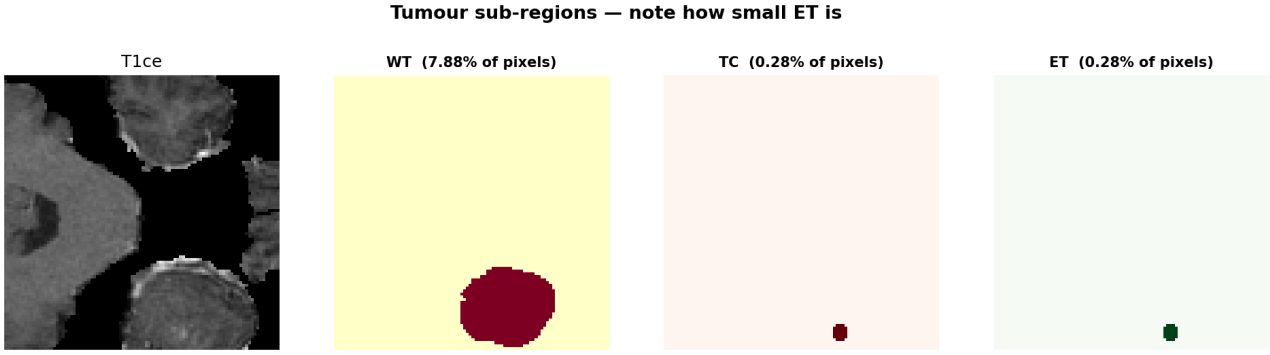


fig09 The nested regions CITE

WT, TC and ET on a real slice. The nesting is the structural claim: these are not three separate labels but three shells, so a voxel inside ET is also inside TC and WT and accumulates all three weights.

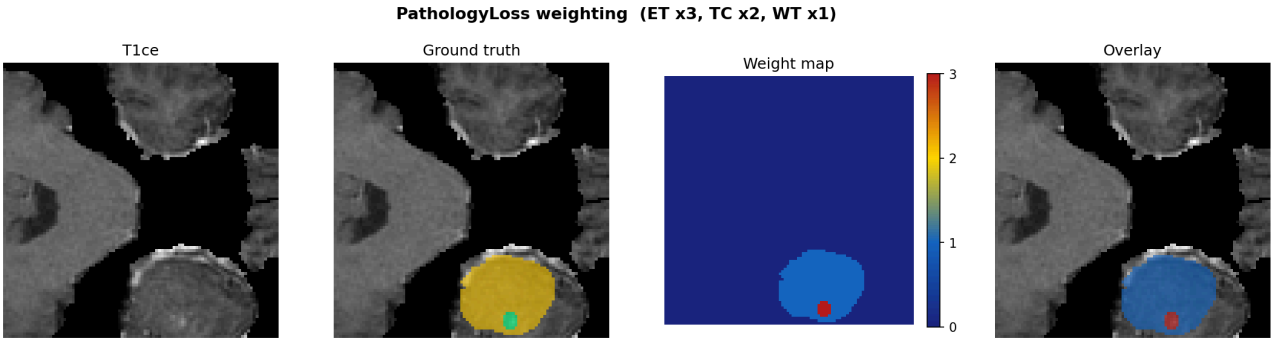


fig11 The weight map CITE

The coefficient each voxel actually receives. Because every region is normalised by its own area, the multiplier is size-dependent rather than a fixed 6×; measured on real data the effective ratio comes out 2.97 : 2.04 : 1.00, so the intended ordering survives.

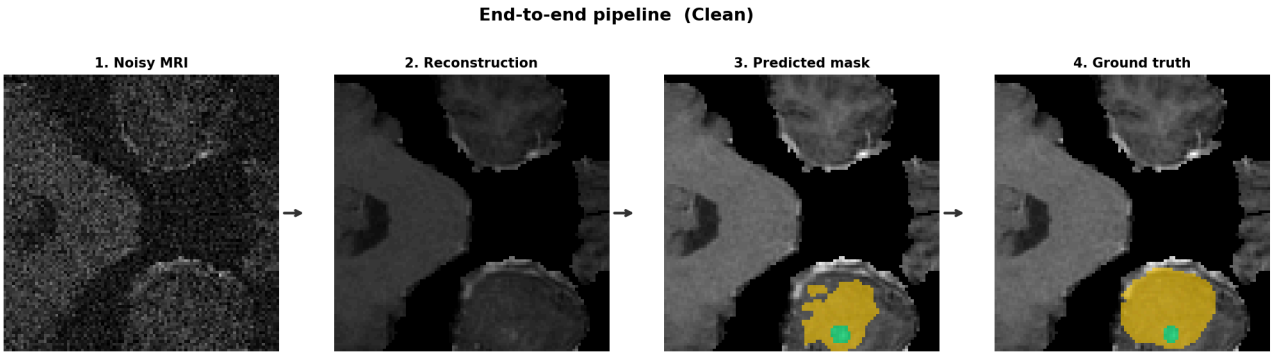


fig13 Evaluation pipeline CITE

Corrupt, restore, then segment with a segmentor trained on clean images and frozen. Every objective is judged by the identical frozen network, so a Dice difference can only come from the restoration.

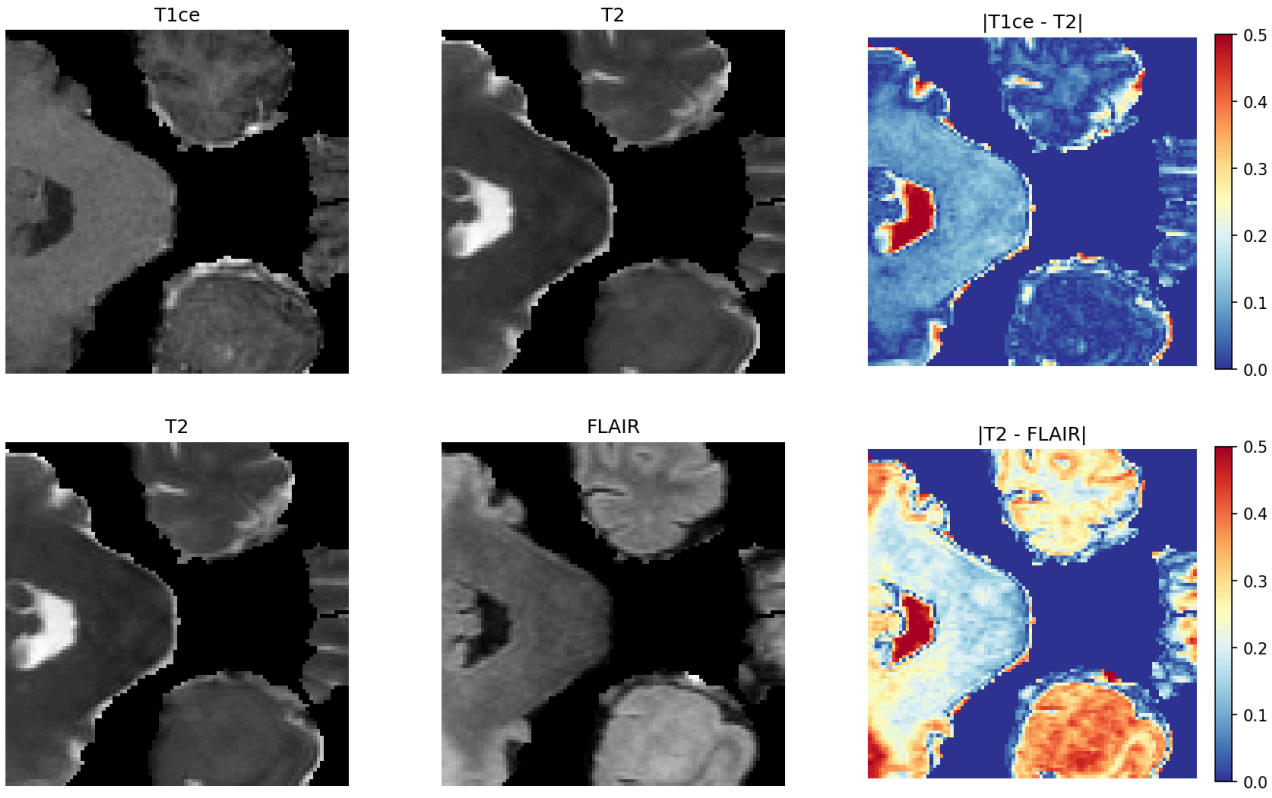


fig14 The cross-modal term CITE

The four modalities are co-registered, so structure should agree across them. This term is in the matched control too — and §5 is where it turns out to be doing most of the work.

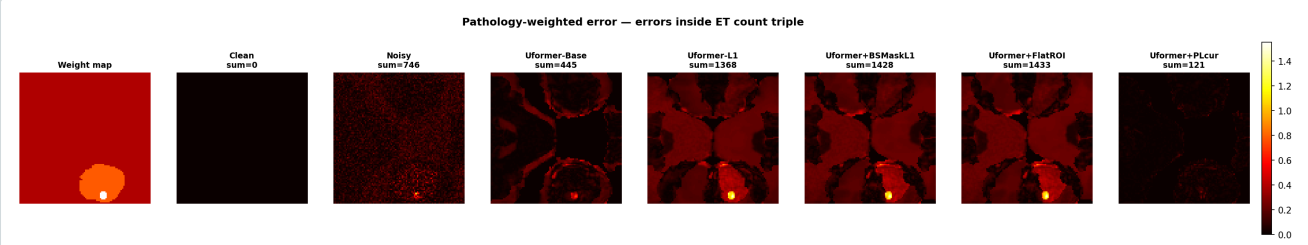


fig12 **Where each objective looks** CITE

Spatial distribution of loss for each objective on one slice. This is the clearest single picture of what distinguishes the arms before any metric is computed.

ACT 03

What came out

All seven arms on slice 95 of BraTS2021_00014, chosen automatically as the slice on which the arms disagree most — a slice they all handle identically demonstrates nothing.

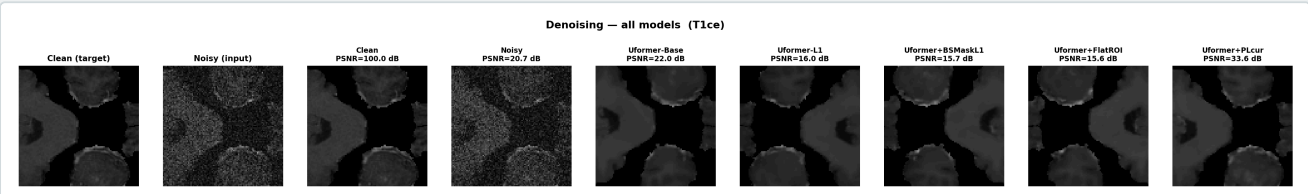


fig03 **Restored output, all arms** CITE

Identical windowing across arms, so brightness differences are real rather than a display artefact. To the eye these are nearly indistinguishable — which is the honest starting point for everything that follows.

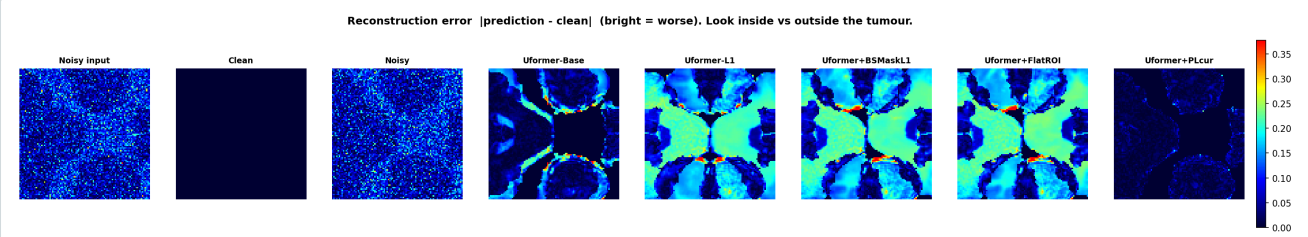


fig04 **Reconstruction error** CITE

|restored − clean| on one shared colour scale, so a brighter panel genuinely means worse. The differences the eye could not find in fig03 are visible here, concentrated exactly where each objective was told to care.

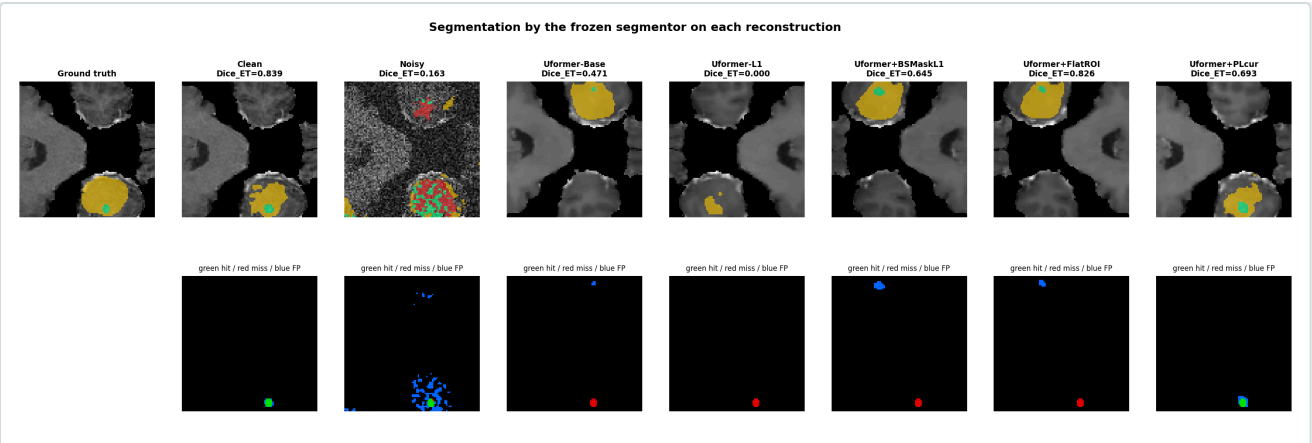


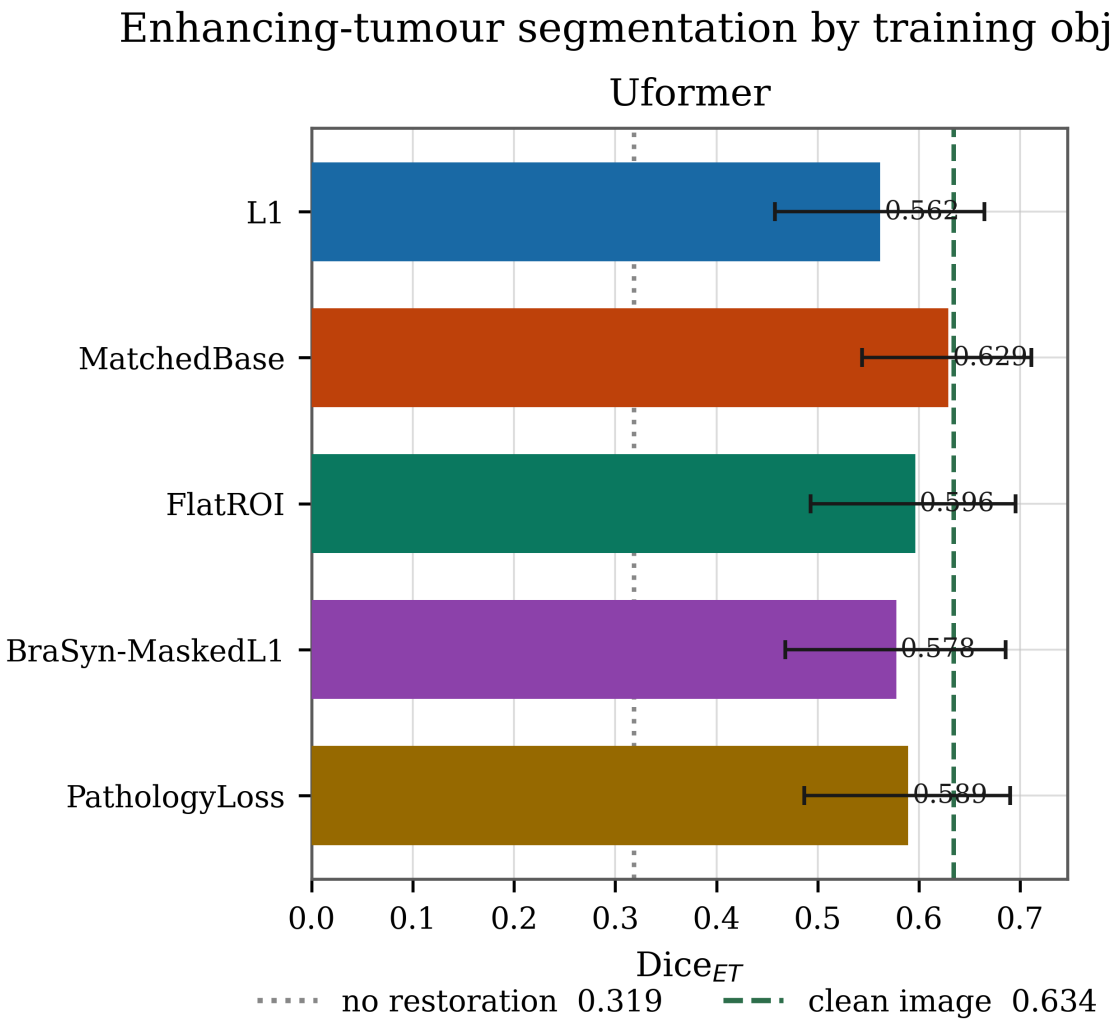
fig05 **Segmentation agreement** CITE

Green found real tumour, red missed it, blue invented it. The blue is the column to read: Dice rewards finding tumour and does not symmetrically punish inventing it.

ACT 04

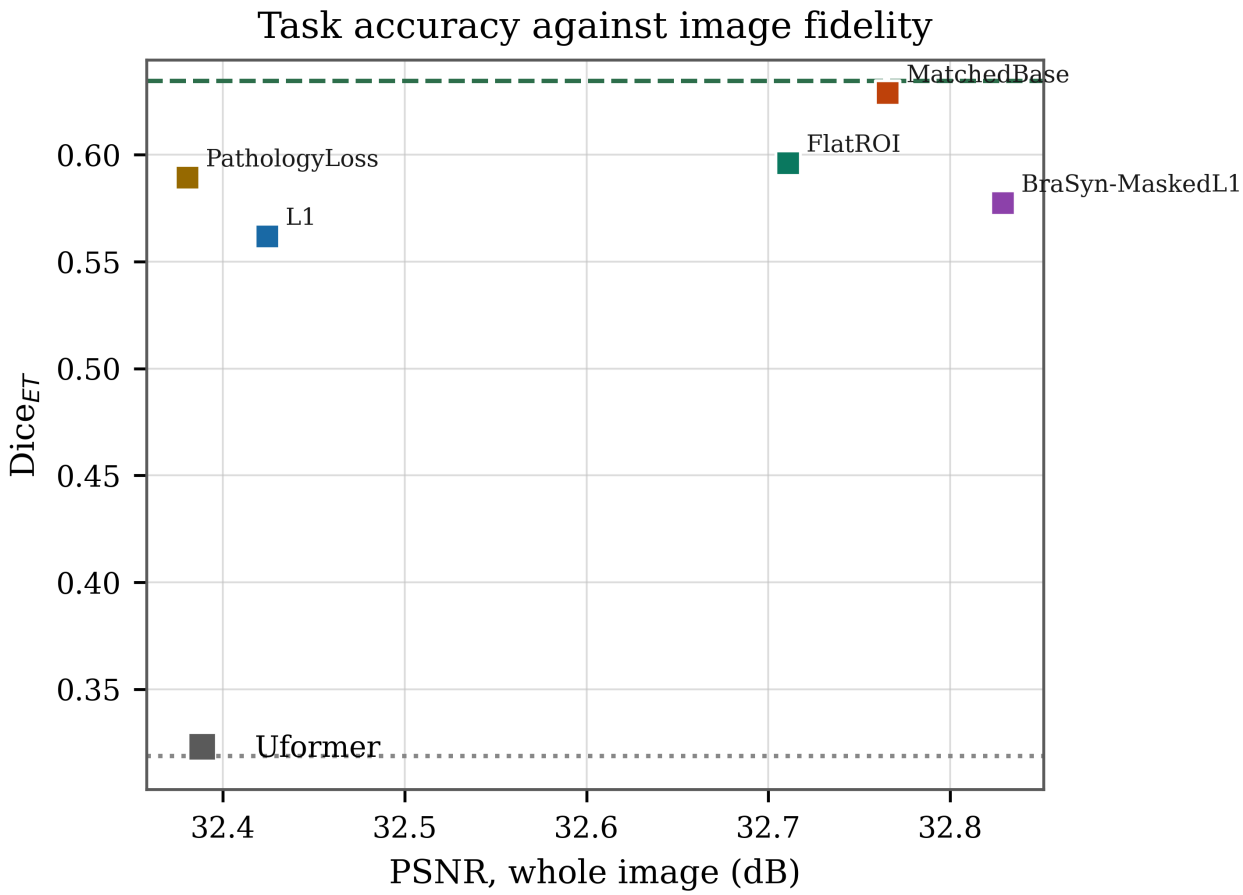
Results

Twenty held-out patients, 1206 slices. Inference is at patient level throughout: 1206 slices from 20 people are not 1206 independent observations, and treating them as such inflates significance roughly tenfold.



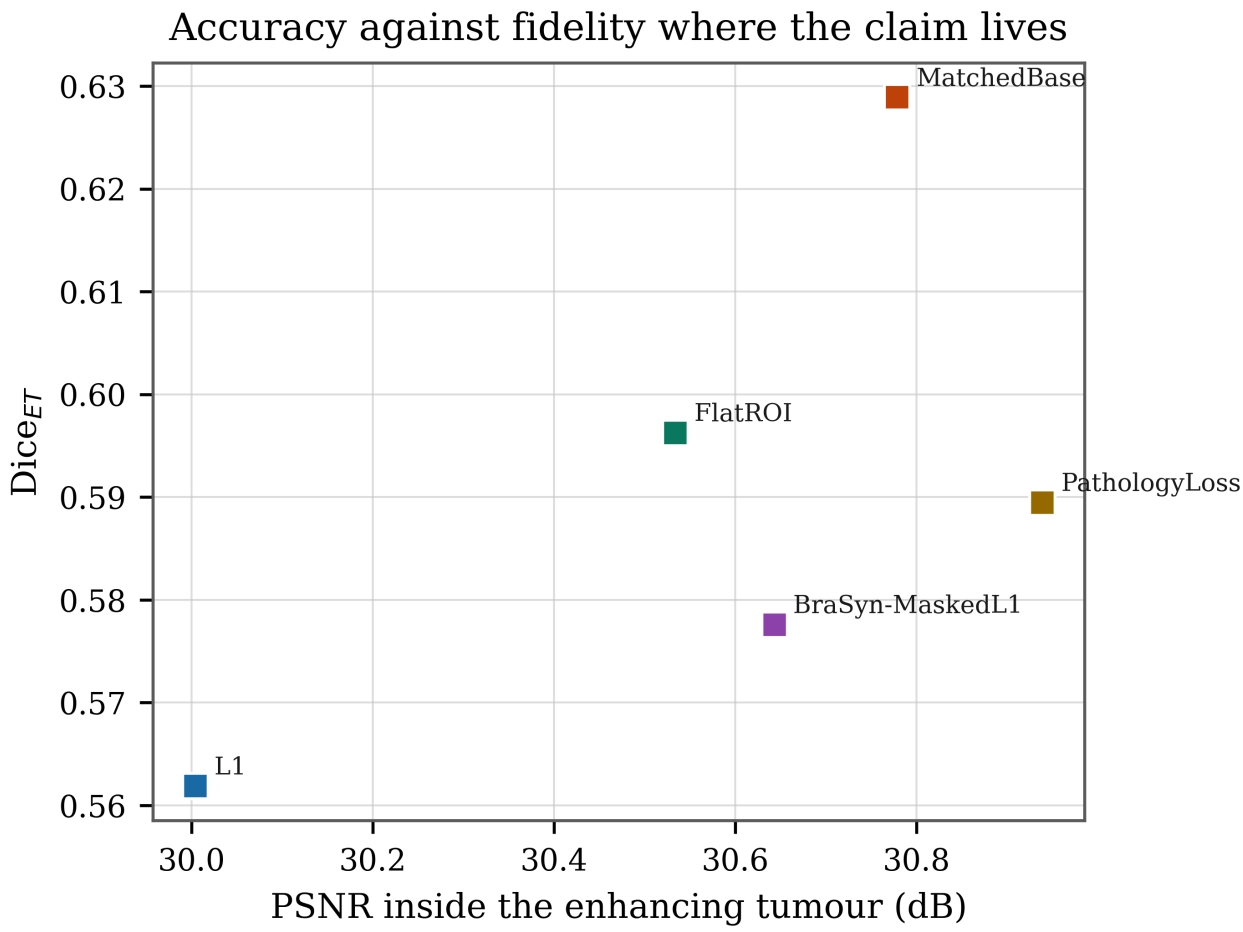
`paper fig1` **Accuracy by objective** CITE

Dice on enhancing tumour per objective, bootstrap 95% intervals over patients, floor and ceiling marked. The matched control — which contains no pathology term at all — sits highest, essentially at the clean-image ceiling.



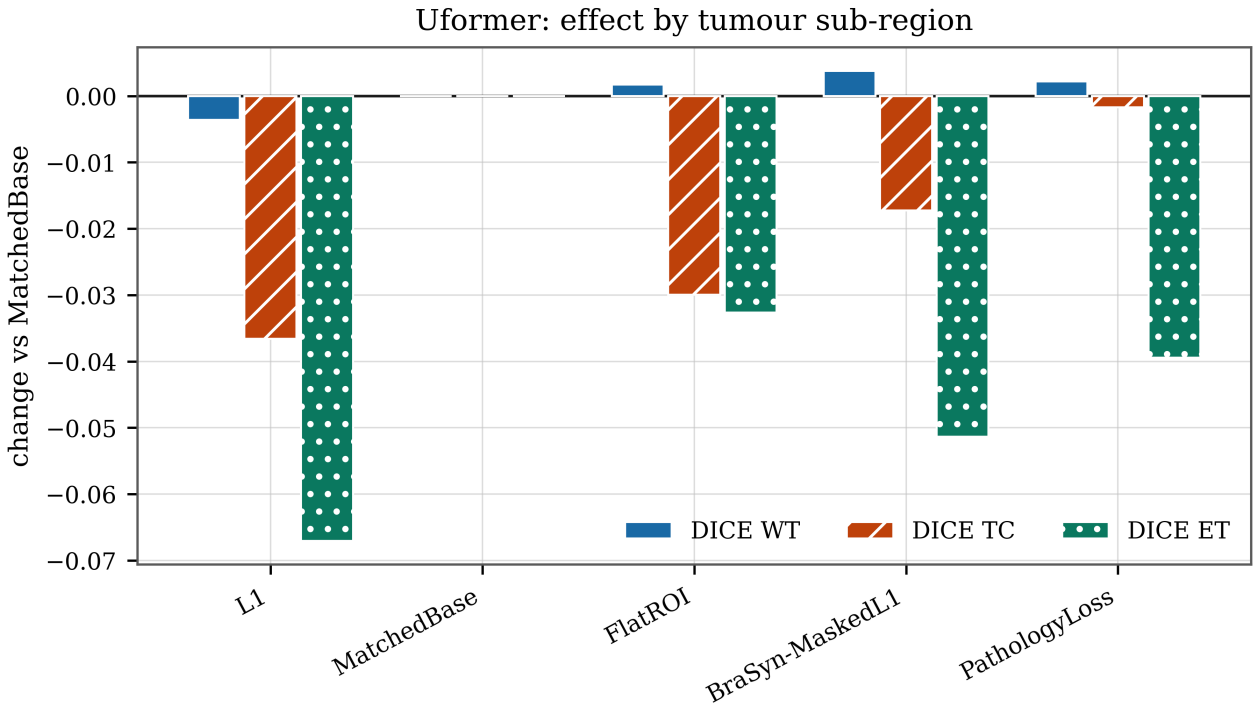
`paper fig2` **Accuracy against global fidelity** CITE

The arm with the best picture is not the arm best for the tumour. PathologyLoss has the *lowest* global PSNR of the five trained arms, by design.



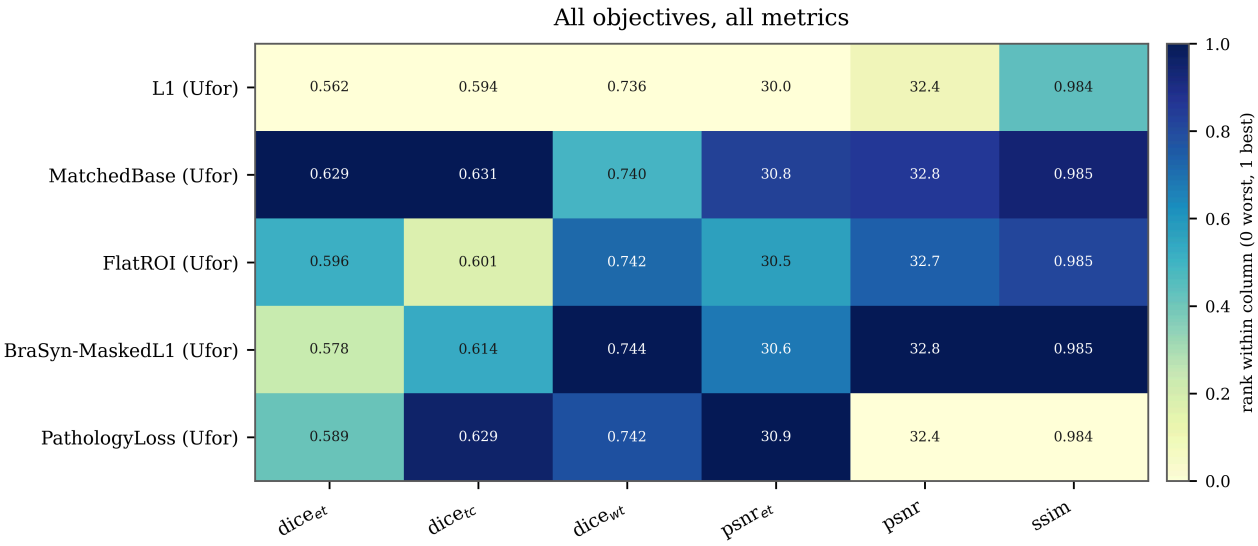
`paper fig3` **Accuracy against targeted fidelity** CITE

The same plot against PSNR measured only inside enhancing tumour — where the claim actually lives. PathologyLoss is furthest right: +0.94 dB over L1 on 19 of 19 patients. This is the run's most robust positive result.



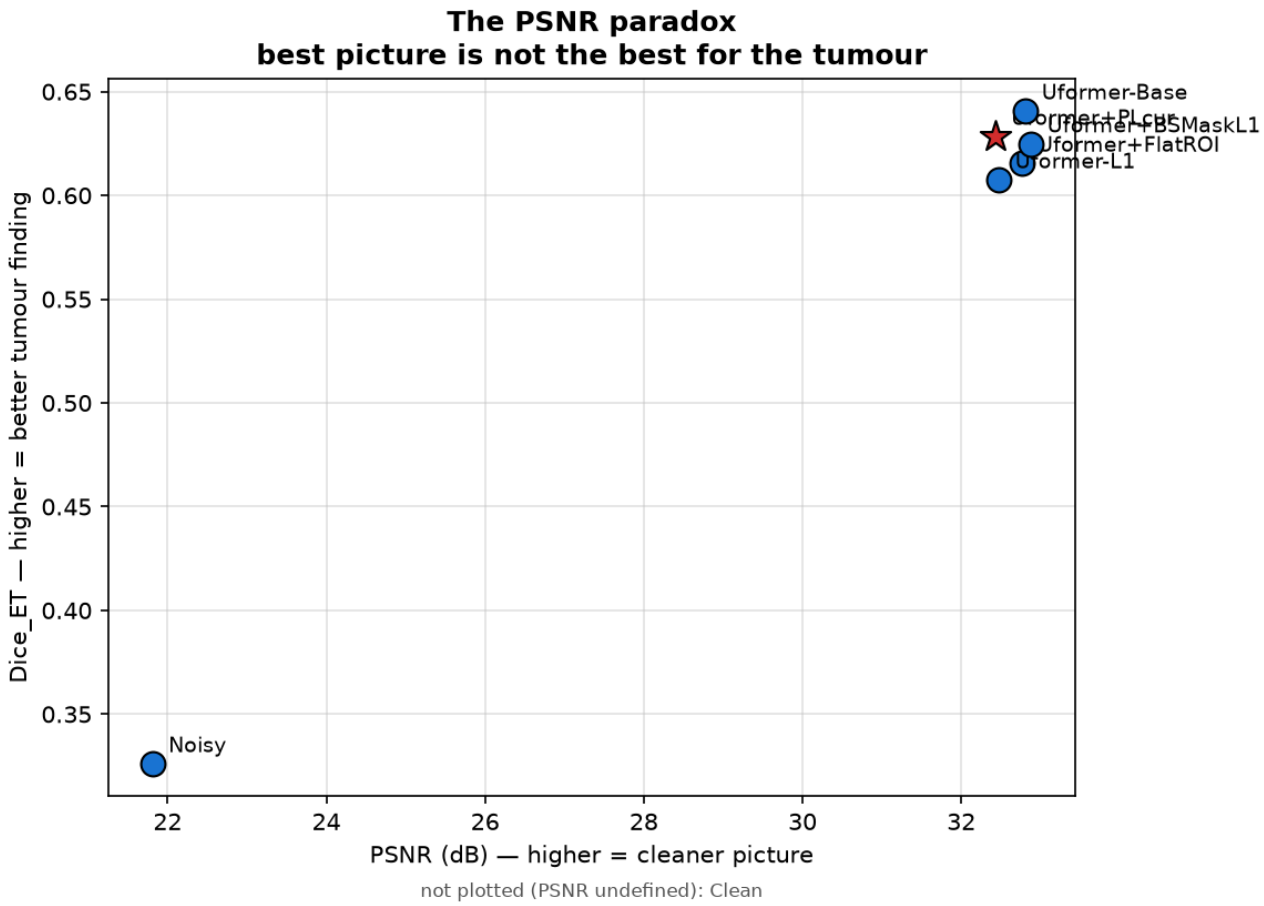
`paper fig6` **Per-region breakdown** CITE

Every arm plotted as its *change from the matched control*, split by region. Read the sign carefully: the bars run downward, so each arm shown is worse than the control. The gaps are largest on enhancing tumour — the innermost, most heavily weighted shell — and near zero on whole tumour. PathologyLoss sits about 0.04 Dice *below* the control on ET, which is this plate's version of the central negative result.



`paper fig4` **Every arm, every metric** CITE

The whole result surface at once. The trade-off reads as a diagonal: arms that win on region-restricted fidelity lose on global fidelity, and no arm wins everywhere.



`fig07` The fidelity paradox

CARE

The run's own version of fig2, regenerated after the crash. The clean-image anchor has infinite PSNR and no position on this axis, so it is noted under the plot rather than drawn. Not submission-ready: the five trained arms land almost on top of one another and their labels overlap illegibly. Use `paper fig2` instead, which plots the same relationship with separated labels.

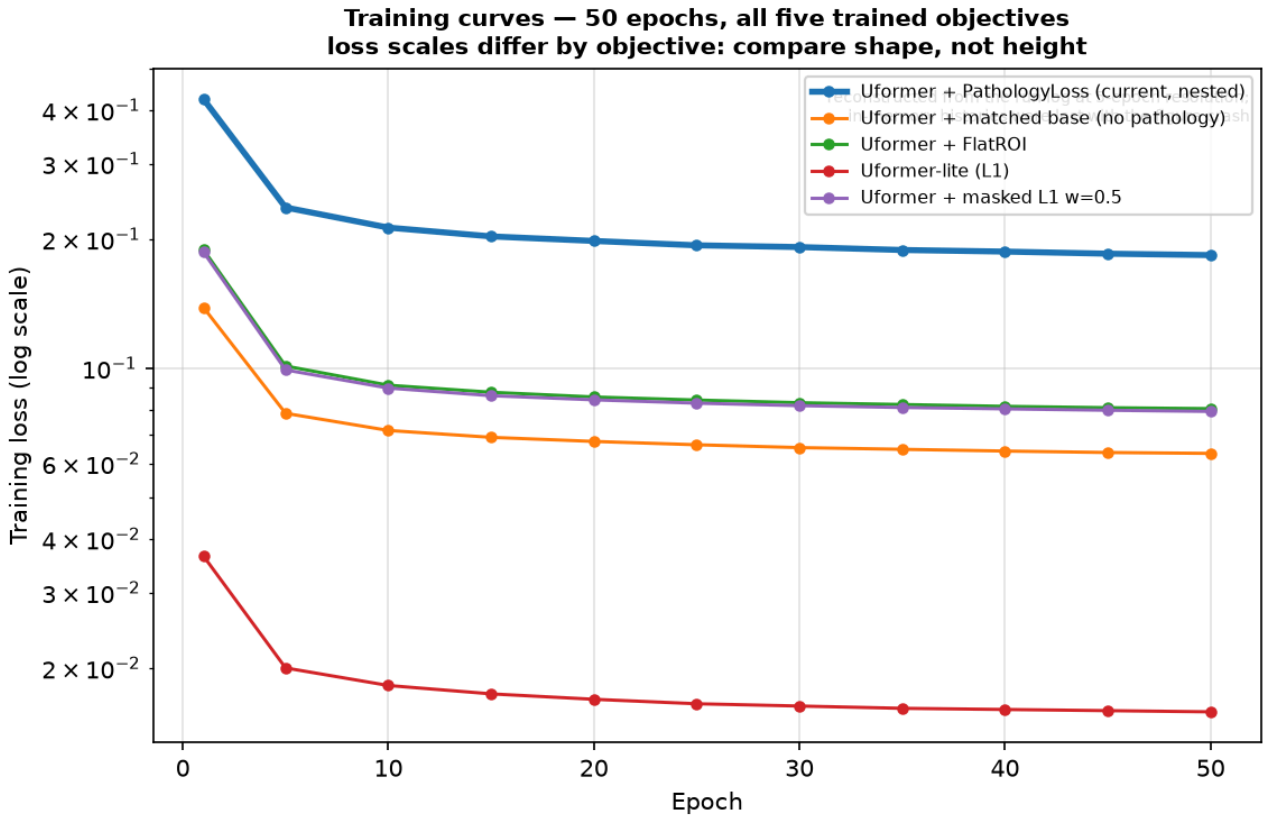


fig08 Training curves CARE

All five objectives, log scale. Loss heights are not comparable — each arm minimises a different functional — so read shape only. Two caveats: the per-epoch histories were held in memory and lost when the run crashed, so this is rebuilt from the log at 5-epoch resolution; and all five were still improving slightly at epoch 50.

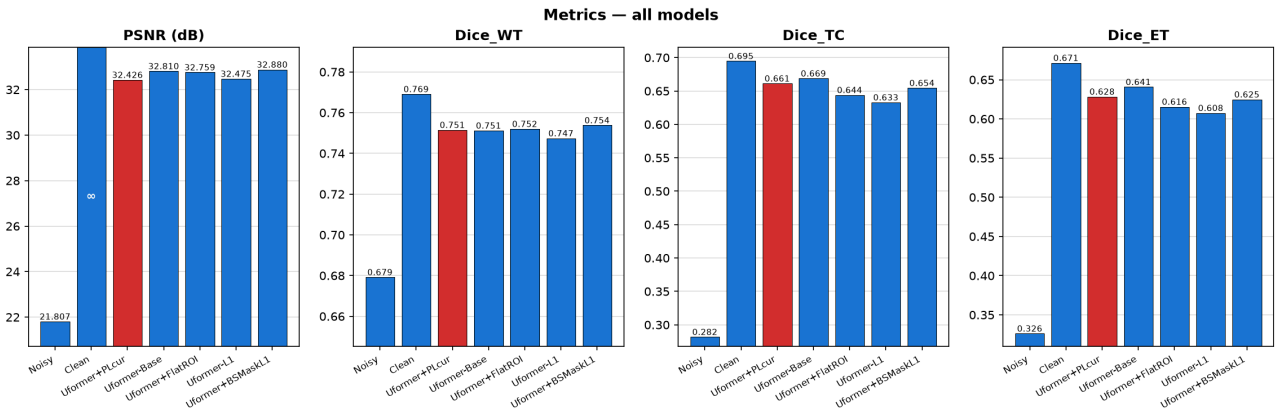


fig06 Metric bars CARE

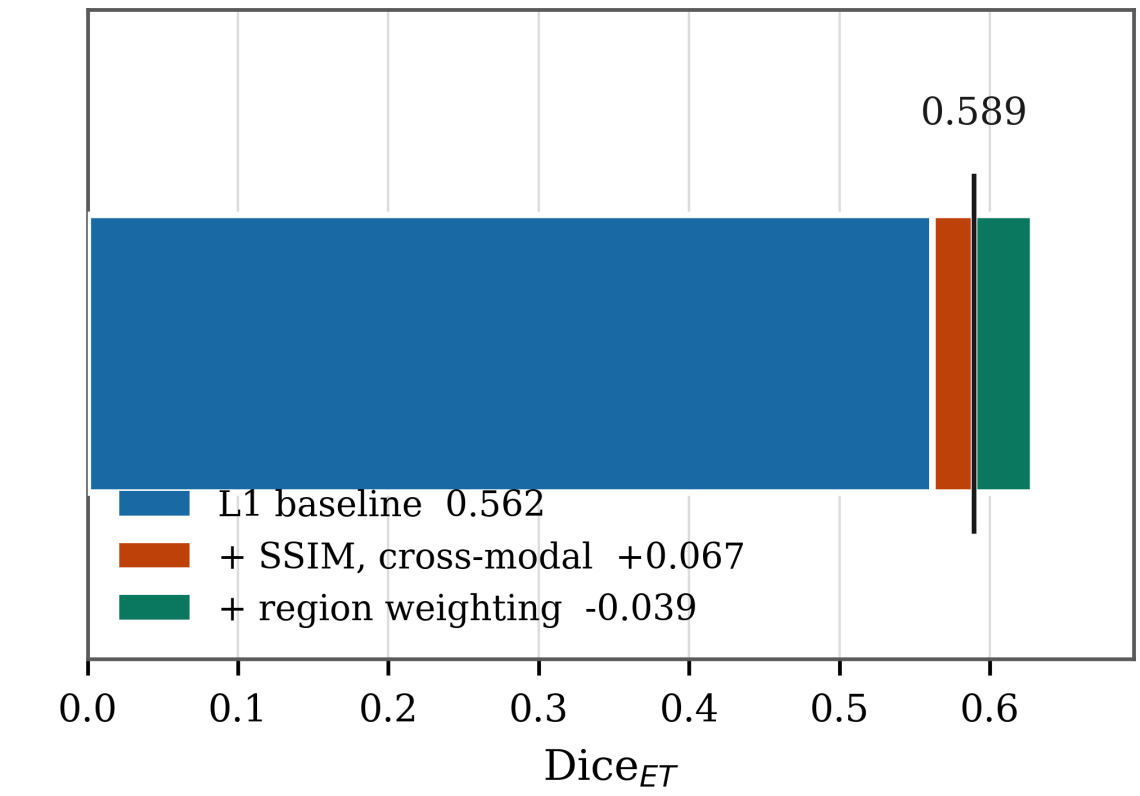
The figure whose crash cost the run its exit code: the clean arm's PSNR is infinite and matplotlib refused the axis. Now drawn from the finite bars with the infinite one marked. Use with care — these are slice-level pooled means, which are descriptive only; paper fig1 is the version with valid intervals.

Results — mask? flags unequal information at inference

Method	PSNR	SSIM	Dice_WT	Dice_TC	Dice_ET	mask?
Noisy	21.81	0.839	0.679	0.282	0.326	-
Clean	inf	1.000	0.769	0.695	0.671	-
Uformer+PLcur	**32.43**	**0.984**	**0.751**	**0.661**	**0.628**	**-**
Uformer-Base	32.81	0.985	0.751	0.669	0.641	-
Uformer+FlatROI	32.76	0.985	0.752	0.644	0.616	-
Uformer-L1	32.47	0.985	0.747	0.633	0.608	-
Uformer+BSMaskL1	32.88	0.985	0.754	0.654	0.625	-

`fig15` **Results table** CARE

Convenient for a slide, but slice-level like fig06. For the manuscript use `table1_main_results.tex`, which is patient-level and carries confidence intervals.



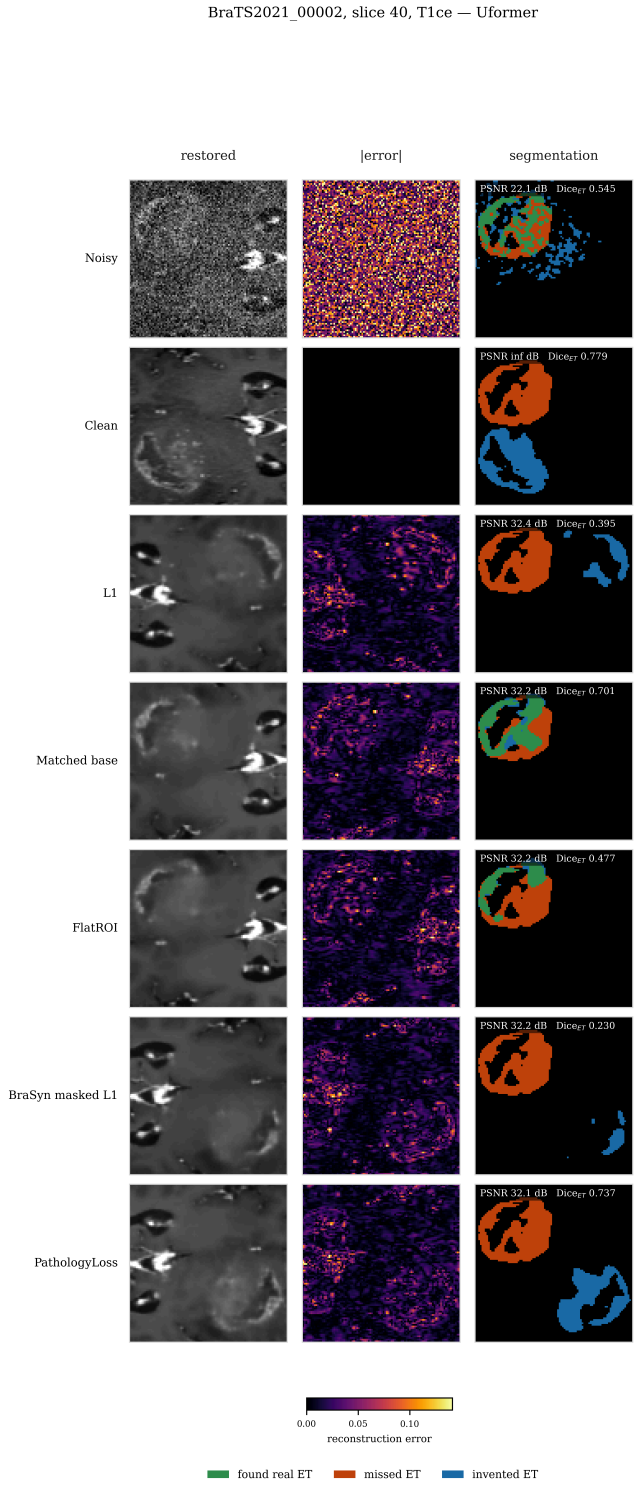
`paper fig5` **Gain decomposition** RETIRE

This plate attributes the improvement 65% to the base terms and 35% to the pathology term. At 50 epochs that split is approximately 100 : 0 — the matched control beats plain L1 by *more* than PathologyLoss does. The figure now contradicts its own source data and should not be shown.

ACT 05

The decisive plate

The figure a reviewer looks at first, and the first one this project could build — earlier runs saved no slice dumps.



`paper fig7` **Qualitative comparison, all seven arms** CITE

BraTS2021_00002, slice 40, T1ce. Three panels per objective: restored, absolute error, and segmentation — green found real enhancing tumour, red missed it, blue invented it.

Read the blue. PathologyLoss carries the highest false-positive rate of any trained arm, 13.1% against 8.8% for plain L1, significantly worse than both L1 and BraSyn. It finds more enhancing tumour and it invents more. No table in the paper shows that; this plate does.

All twenty-two plates come from one run, `phasefinal_seed1`, on 100 BraTS 2021 subjects split 80/20 by patient with zero overlap. Source data is `per_slice_metrics.csv` — 8442 rows, 11 metrics × 1206 held-out slices × 7 arms. Nothing here is simulated or carried over from an earlier run.

One seed. A single-seed result is suggestive, not settled, and 20 patients with Dice standard deviations near 0.24 is thinly powered for the small differences at issue.